EXP 12	Book Issue Workflow

Aim:

To automate the issue of a book: default dates, due date calculation, status and reduction of the available stock by one.

Theory:

The issue process uses four workflows on the Book Issues form:

- **Default Issue and Due Date (on load):** Issue Date = today, Due Date = today + 14 days, Status = Issued.
- **Calculate Due Date (user input of Issue Date):** Recalculates Due Date = Issue Date + 14 days.
- **Validate Book Issue (on validate):** Business rules from Experiment 10.
- **Reduce Stock on Issue (on success):** Available Copies of the book is reduced by exactly 1 and the status is set to Issued, or Overdue when the due date has already passed.

Procedure:

1. Create the workflow Default Issue and Due Date (Book Issues, Created, Load of the form) and add the script.
2. Create Calculate Due Date (Created or Edited, User input of a field, field Issue Date).
3. Create Reduce Stock on Issue (Created, Successful form submission) with a Deluge script action.
4. Open the live Book Issues form, select Divya M and Atomic Habits and submit.
5. Check the Book Catalogue: Available Copies of Atomic Habits falls from 2 to 1.

Program (Default Issue and Due Date / Calculate Due Date):

```
// Book Issues - on load (Created): default issue date and 14-day due date
input.Issue_Date = zoho.currentdate;
input.Due_Date = zoho.currentdate.addDay(14);
input.Issue_Status = "Issued";

// Book Issues - on user input of Issue Date: due date = issue date + 14 days
if(input.Issue_Date != null)
{
    input.Due_Date = input.Issue_Date.addDay(14);
}
```

Program (Reduce Stock on Issue):

```
// Book Issues - after a successful issue (Created): reduce stock by exactly one
bk = Books[ID == input.Book];
bk.Available_Copies = bk.Available_Copies - 1;
if(input.Due_Date == null)
{
    input.Due_Date = input.Issue_Date.addDay(14);
}
if(input.Due_Date < zoho.currentdate)
{
    input.Issue_Status = "Overdue";
}
```

```
else
{
    input.Issue_Status = "Issued";
}
```

Output:

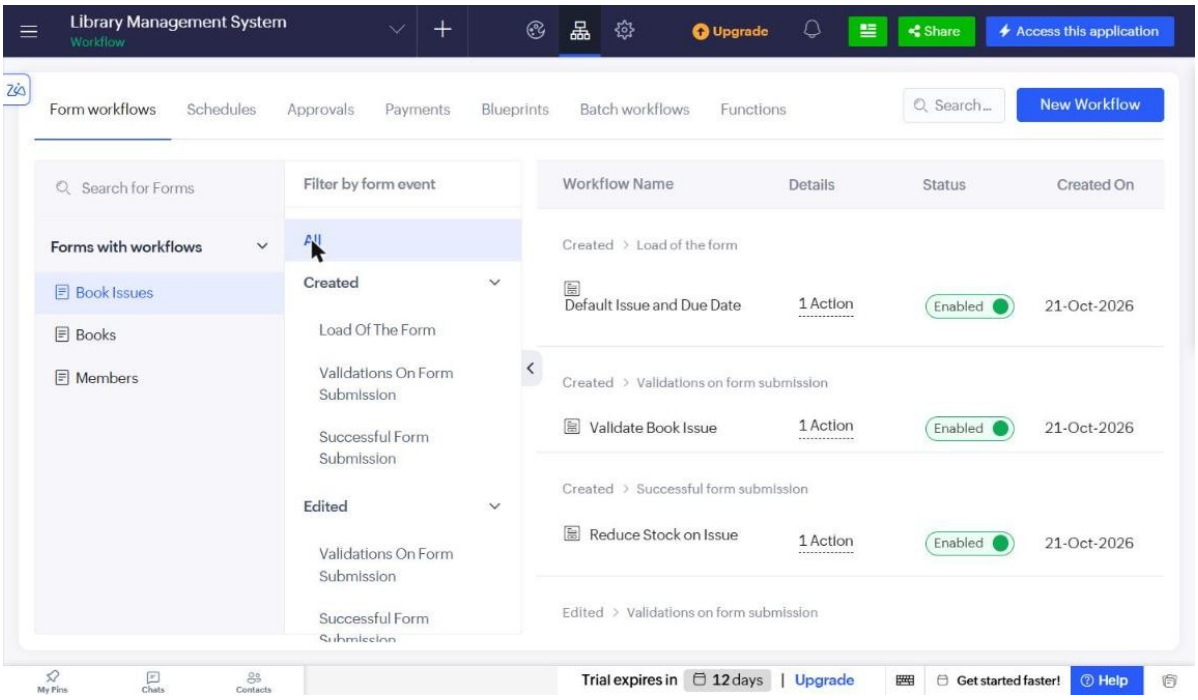


Fig. 12.1: Workflows on the Book Issues form

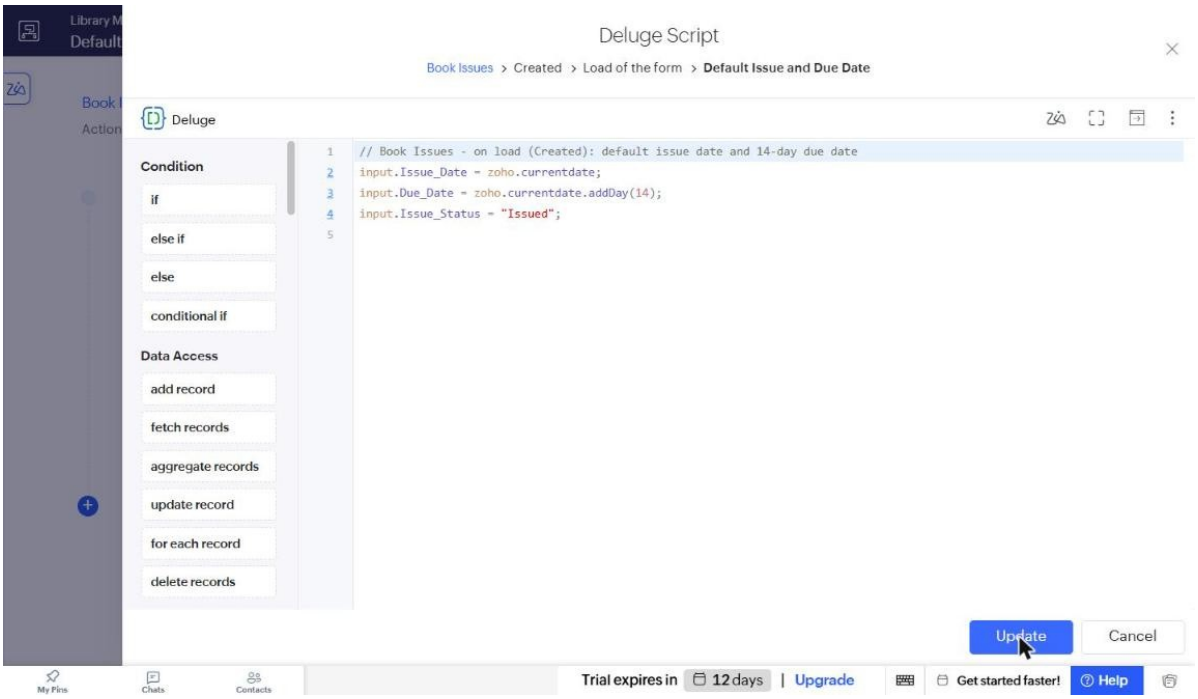


Fig. 12.2: On-load script

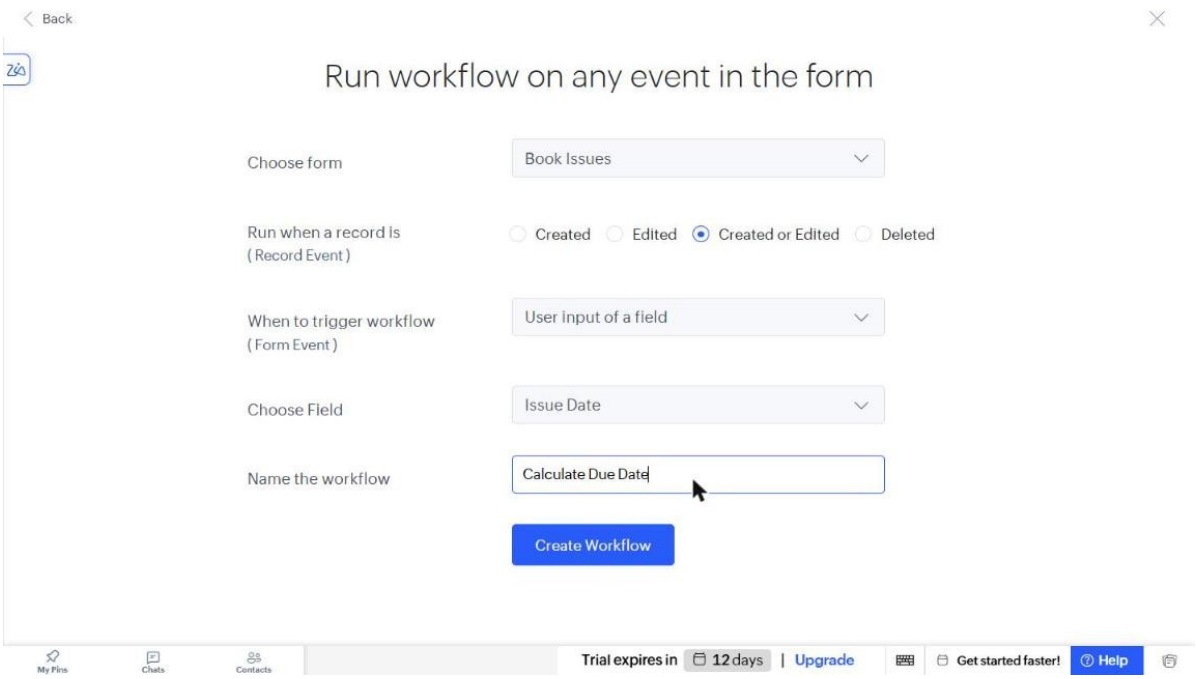


Fig. 12.3: Calculate Due Date workflow

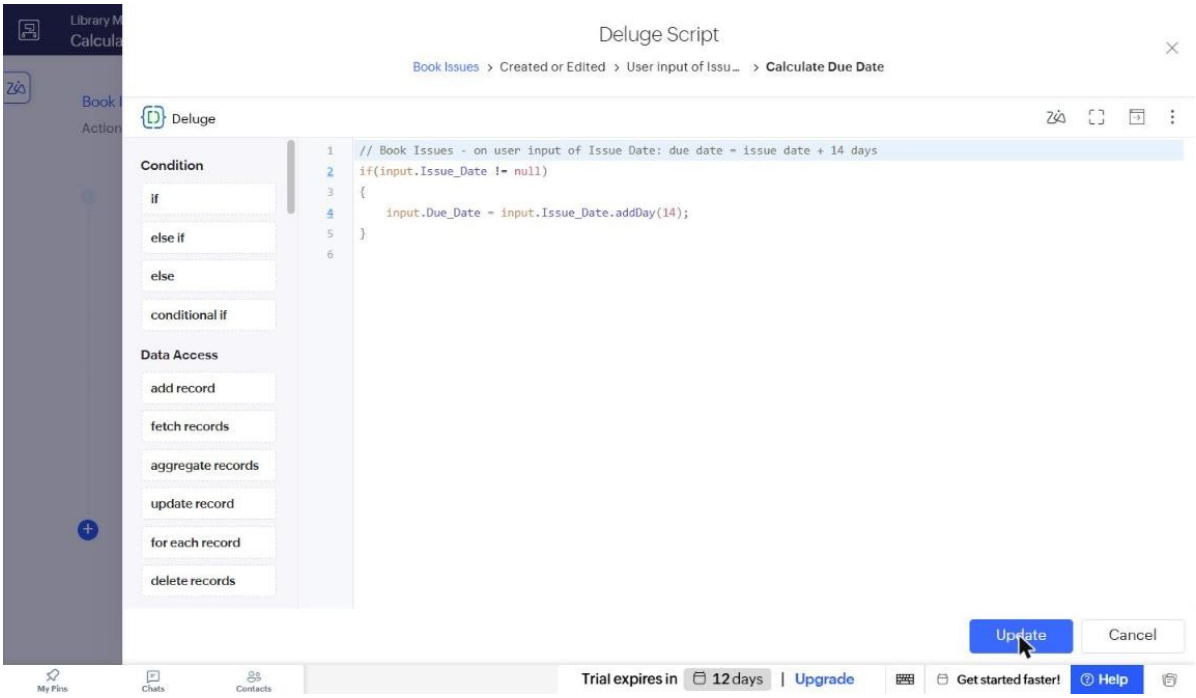


Fig. 12.4: Calculate Due Date script

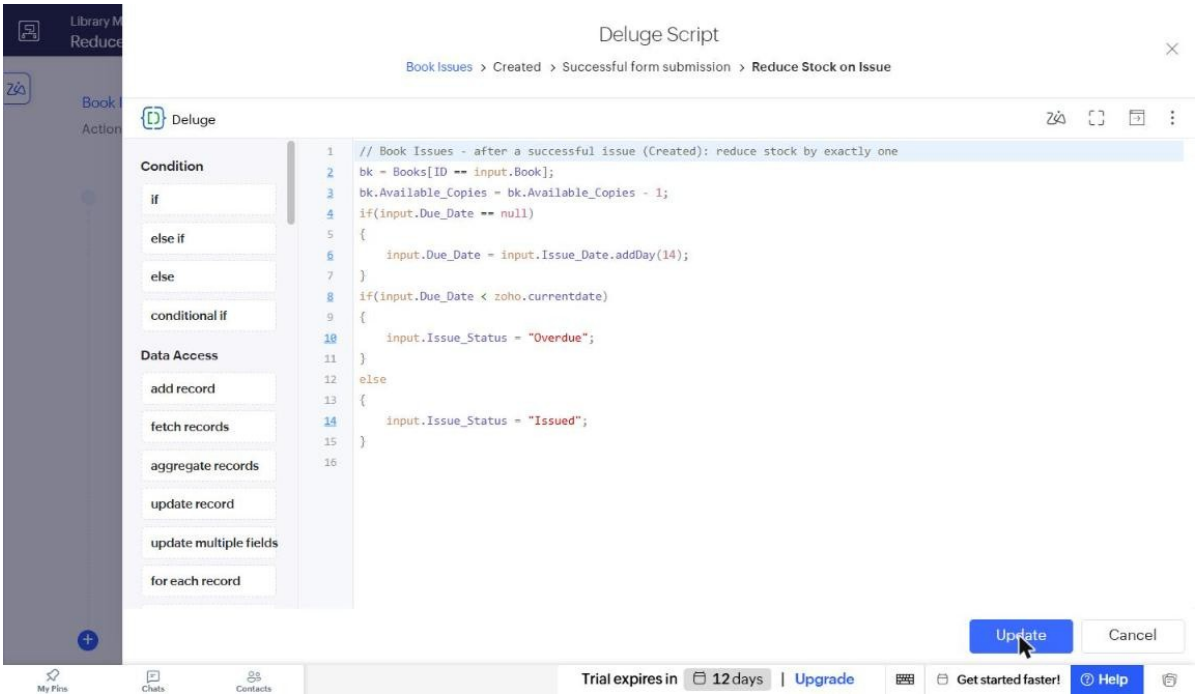


Fig. 12.5: Reduce Stock on Issue script

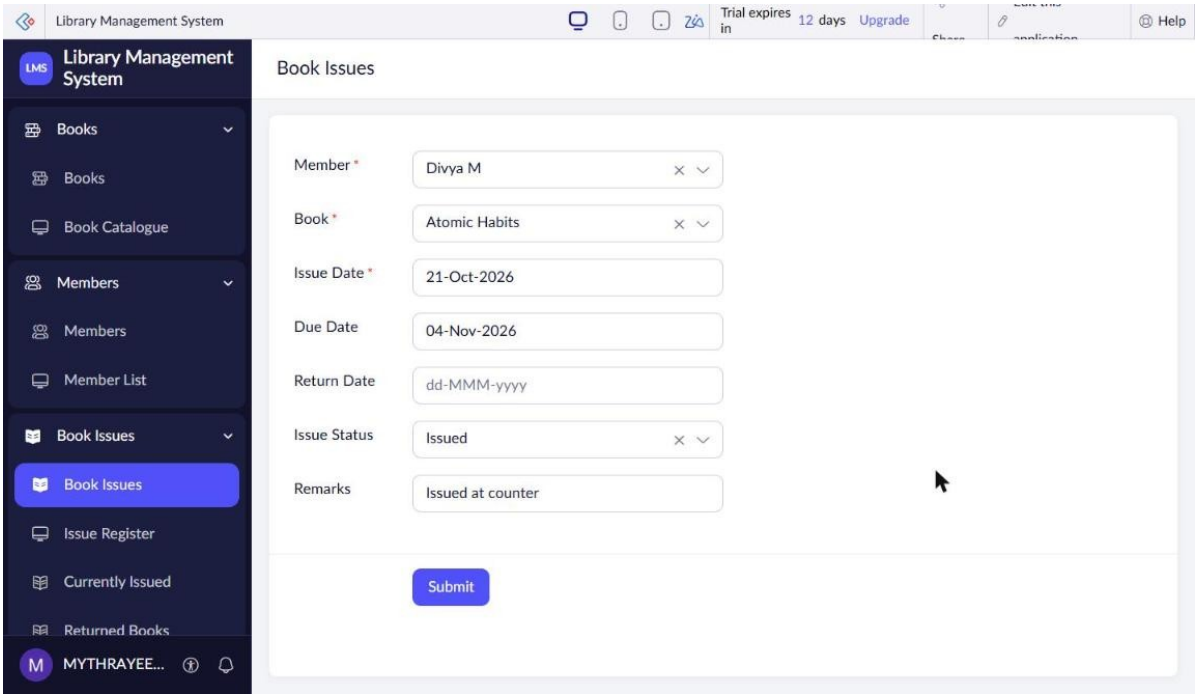


Fig. 12.6: Issue form filled

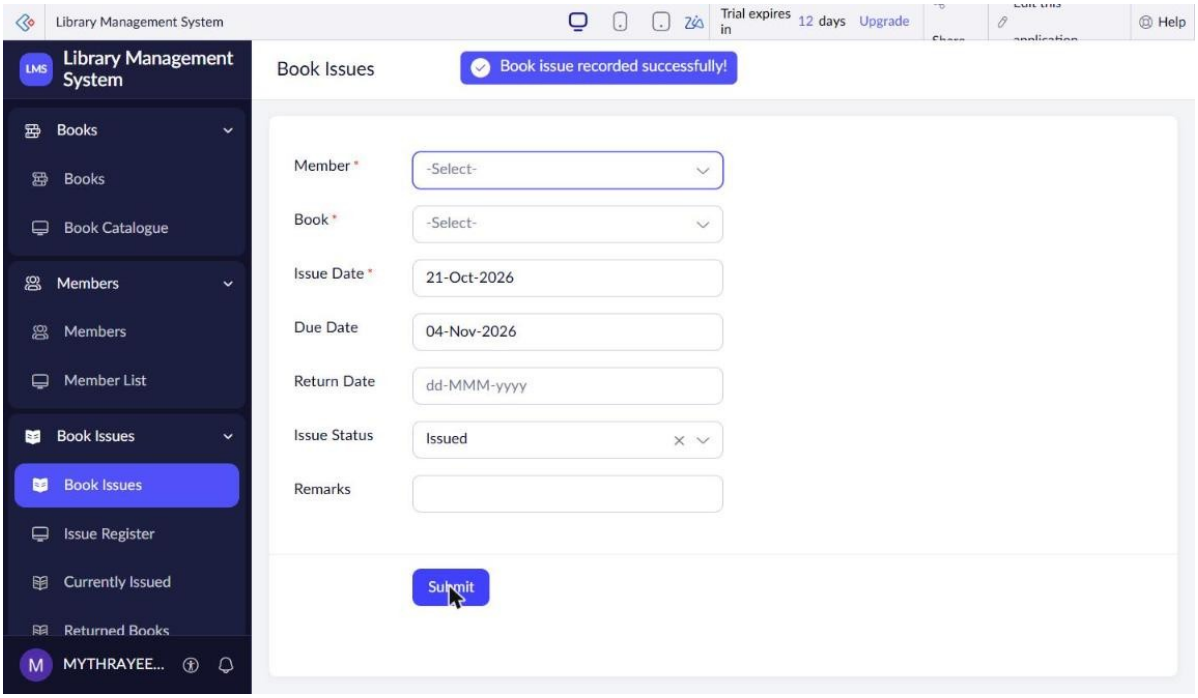


Fig. 12.7: Issue recorded successfully

Observation:

Book	Available before	Issued to	Available after
Clean Code	3	Arun Kumar	2
Sapiens	2	Priya S	1
1984	1	Karthik R	0
Atomic Habits	2	Divya M	1
Astrophysics for People in a Hurry	1	Sneha P	0

Result:

The issue workflow fills the dates, calculates the 14-day due date, sets the status and reduces the available copies by exactly one for every issue.